

Towards a re-decentralization of the Internet

Exploring current Technologies and Initiatives

WORLDLINE

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- Since 1999
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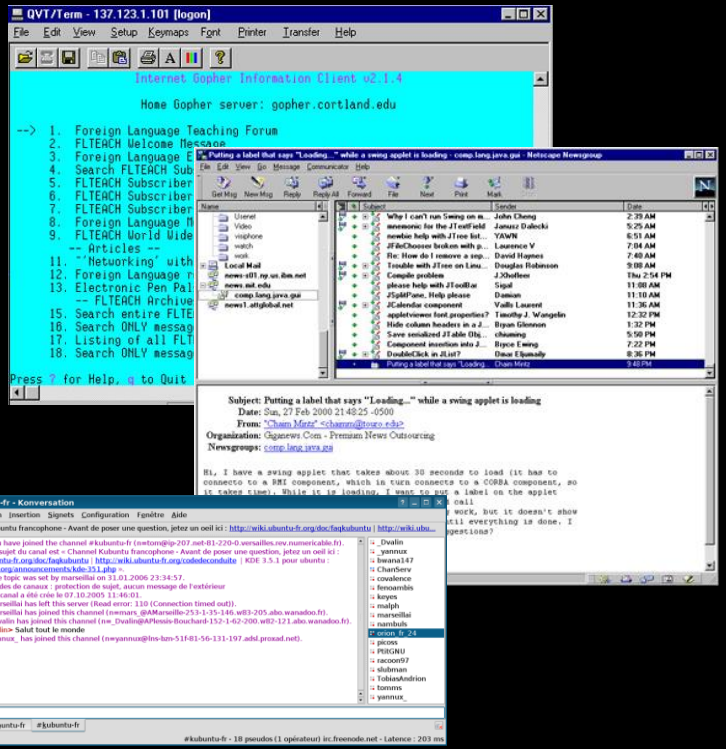


Get in touch

@RaphaelSemeteys
mastodon.social/@raphiki

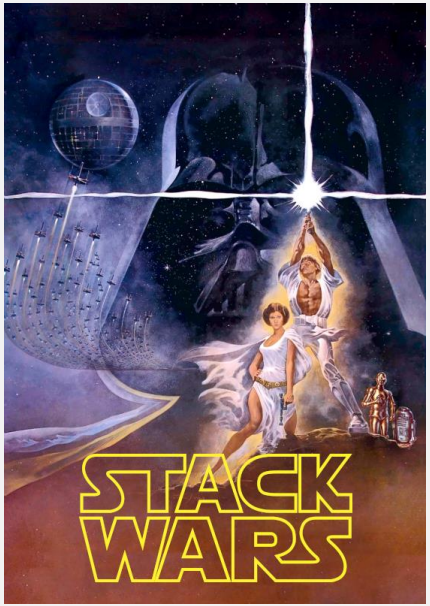
blog.worldline.tech/author/raphaël-semeteys.html
dev.to/raphiki/

linkedin.com/in/raphaelsemeteys/


QVT/Term - 137.123.1.101 [logon]
File Edit View Setup Keymaps Font Printer Transfer Help
Internet Gopher Information Client v2.1.4
Home Gopher server: gopher.cortland.edu
1. Foreign Language Teaching Forum
2. FLTEACH Welcome Message
3. Foreign Language E
4. Search FLTEACH Sub
5. FLTEACH Subscriber
6. FLTEACH Subscriber
7. FLTEACH Subscriber
8. Foreign Language M
9. FLTEACH World Wide
-- Articles --
11. "Networking" with
12. Foreign Language n
13. Electronic Pen Pal
-- FLTEACH Archive
15. Search entire FLTE
16. Search ONLY messag
17. Listing of all FLT
18. Search ONLY messag
Press ? for Help, q to Quit
Putting a label that says "Loading..." while a swing applet is loading - comp.lang.java.gui - Netscape Newsgroup
Subject: Putting a label that says "Loading..." while a swing applet is loading
Date: Sun, 27 Feb 2000 21:48:25 -0500
From: "Chaim Mintz" <chaimm@tsoro.edu>
Organization: Ogranews Com - Premium News Outsourcing
Newsgroups: comp.lang.java.gui
Hi, I have a swing applet that takes about 30 seconds to load (it has to connecto to a RMI component, which in turn connects to a CORBA component, so it takes time). While it is loading, I want to put a label on the applet
#kubuntu-fr - Konversation
Fichier Edition Insertion Signets Configuration Fenêtre Aide


Instant Messaging
Cab Hailing
Shopping
SUPER APP
Delivery
Banking and payments
Social Media
Inverstment

Me



Decentralization (infrastructure)
Collective creation of the Internet

My Kids



Centralization (business rules/value)
Rise of social/business platforms

Tomorrow?



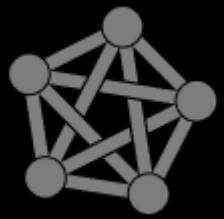
Decentralization (business rules/value)
New Decentralized Protocols

Decentralization





Web3
Web 3.0
Fediverse
AI

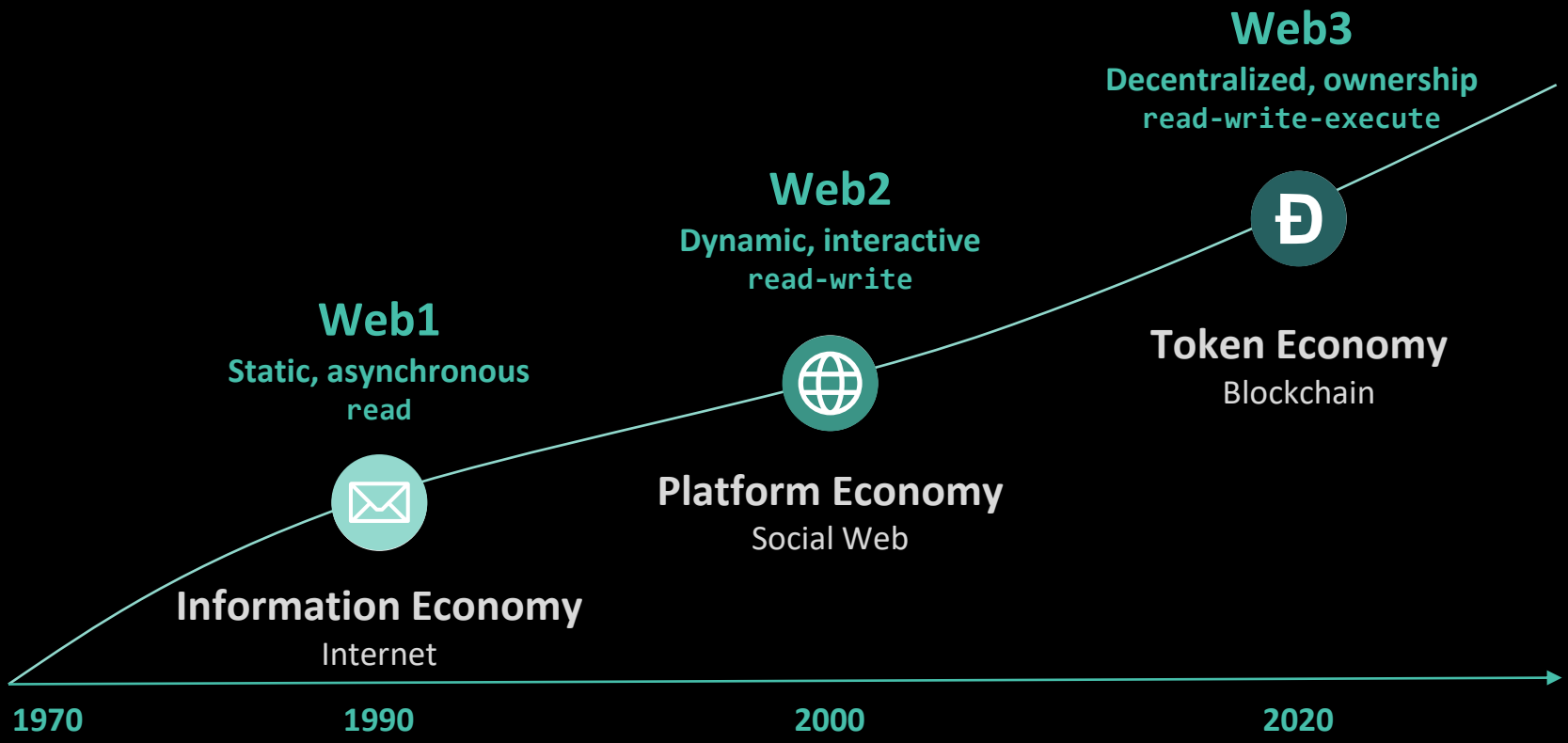




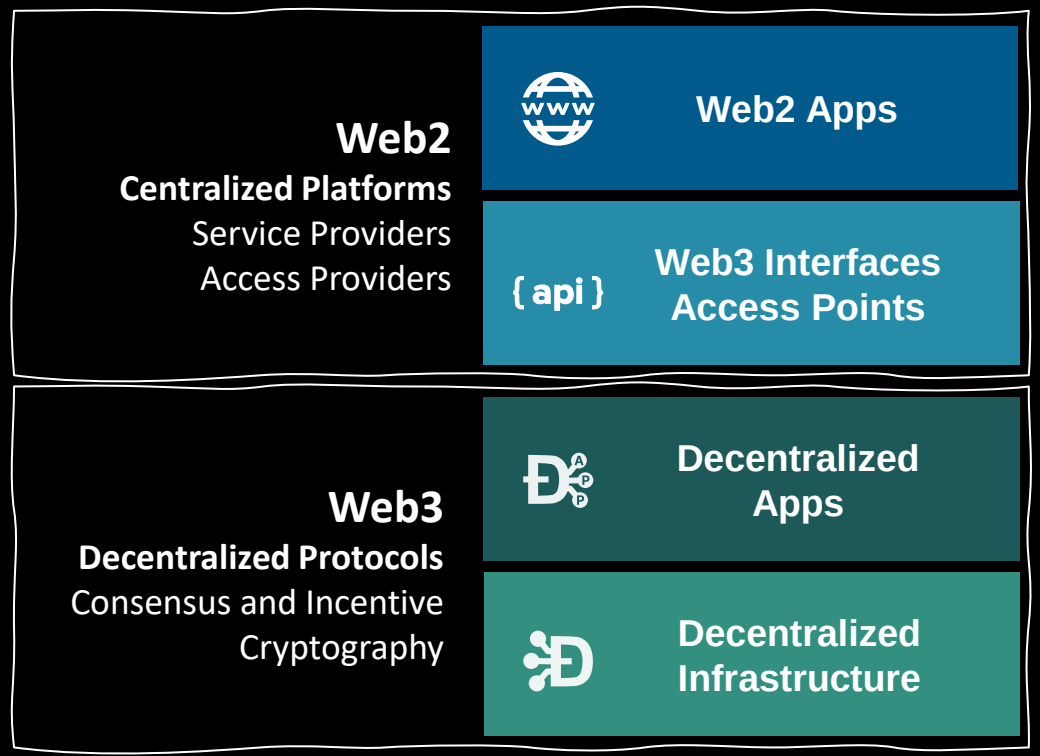


Web3

Blockchain & Web3



Web3 Stack



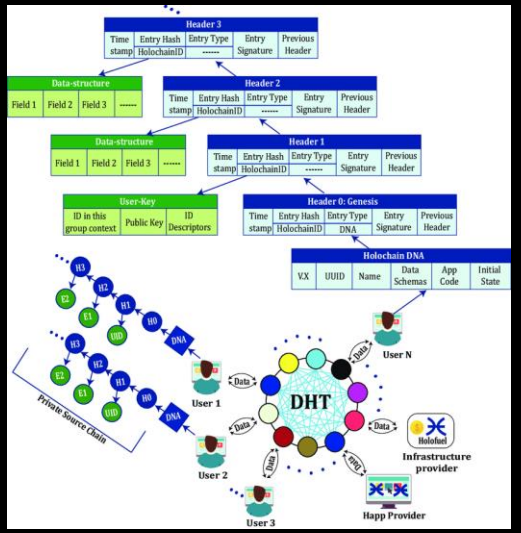
End-user frontends
Wallets

Interaction with decentralized systems
APIs
Oracles

Code is Law
Smart Contracts
Specific tokens (basic, NFT, DAO, etc.)

Trust and value at protocol level
Decentralized Networking/Execution/Storage
Native tokens (coins)

- **Agent-Centric Architecture**
  - Each user (agent) has their own personal Holochain
  - P2P Application Framework
- **Distributed Hash Table (DHT)**
  - Database distributed across many nodes
  - Key-value system: data indexed and accessed via unique keys
- **No global consensus**
  - Distributed Validation
  - Each node validates its own transactions according to predefined rules
  - No native token
- **Decentralized applications (hApps)**
  - Social: Kizuna (IM), Flux (collaboration), Neighborhoods (DAO)
  - But also: acorn (project management), hREA (economic coordination)



Source: Thinking Out of the Blocks
Holochain for Distributed Security in IoT Healthcare



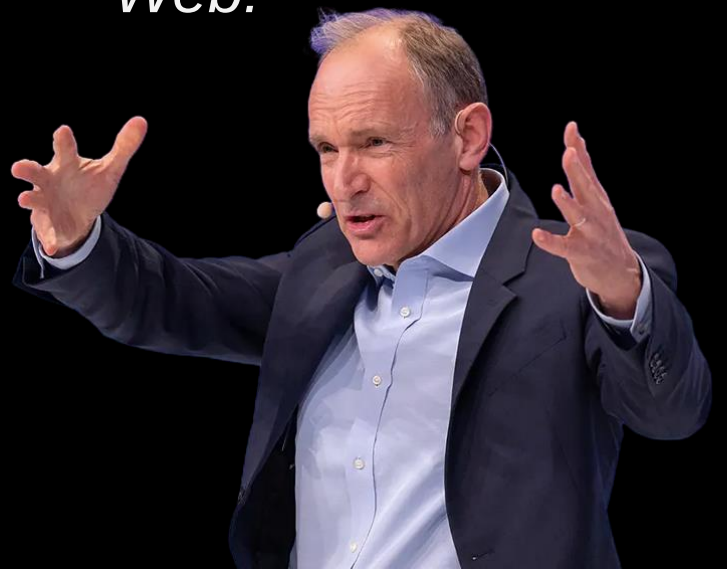
Web 3.0

Semantic Web & Web 3.0

- **Semantic Web: Data Intelligence**
  - Original idea of Tim Berners-Lee
  - Data also interpreted by machines
  - Rich and meaningful links

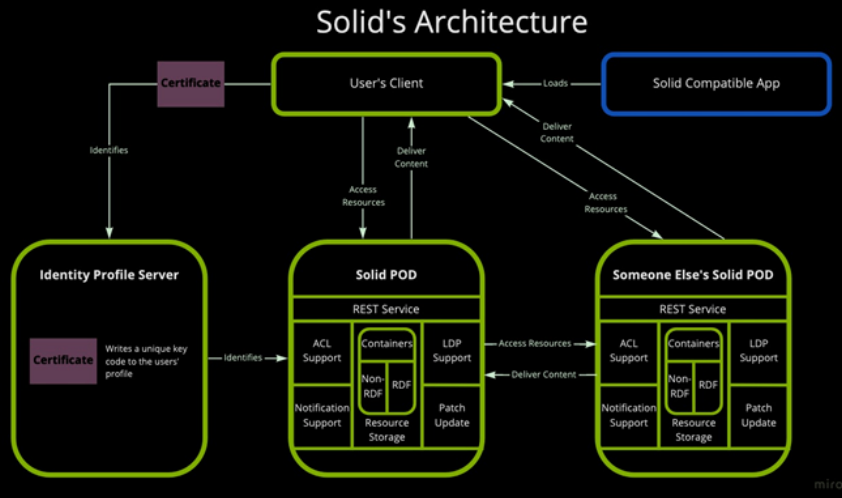
- **Web 3.0: for a Smarter Internet**
  - Not to be confused with the previous misnamed Web3!
  - Part of the re-decentralization movement
  - Focuses on personal data and ownership
    - Data decentralization
    - Interoperability and open standards
    - Security & Privacy

Let's fix the Web!



Solid

- **Solid = « Social Linked Data »**
  - Open source project initiated by Tim Berners-Lee
  - MIT then W3C
- **Web 3.0 concrete implementation**
  - Decentralization: user-chosen storage
  - Security: authentication and access control
  - Interoperability: open Web protocols and standards
- **Technical aspects**
  - Pods (Personal Online Data Stores)
  - Solid Servers (Pod providers or self-hosted)
  - Data: RDF or not, JSON-LD storage
  - WebID, Solid-OIDC, ACL



Source: solid.georgetown.domains



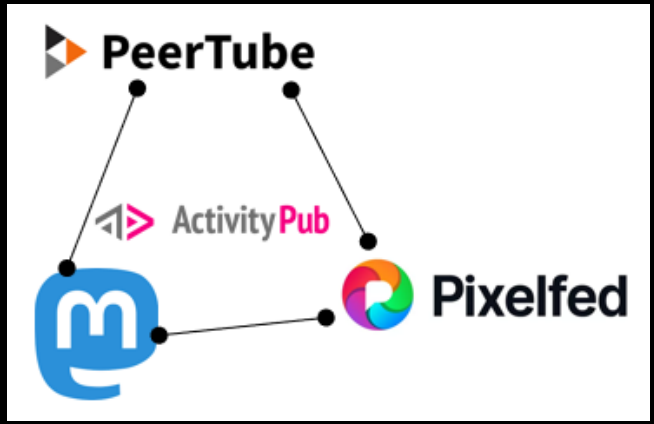
Fediverse

Fediverse

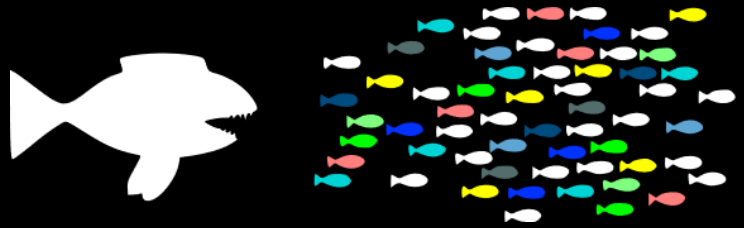
Decentralization through federation of social and collaborative platforms

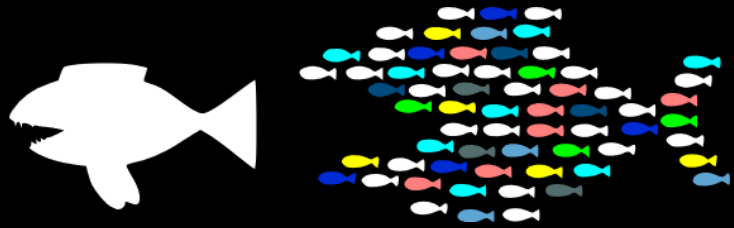
- **Interoperability Protocols**
  - ActivityPub: W3C standard, content and notification management
  - OStatus, Diaspora Protocol, Matrix Protocol
  - WebTorrent: P2P content delivery
- **Size doesn't matter**
  - Personal nodes or hosting services
  - Opensource implementations
- **Examples**
  - Microblogs (Mastodon, Pleroma, Write Freely)
  - Images (PixelFed) & Videos (PeerTube)
  - Chat (Matrix protocol and its gateways)
  - And also, Books (BookWyrm), Music (Funkwhale)...
  - ActivityPub Wordpress plugin

POPULATION	~ 12 019 333	accounts
ACTIVE USERS	~ 3 777 905	people
SERVERS	> 19 672	instances



Fediverse





Beyond cloning centralized platforms

- Returning to the notion of network and mesh
- Return of protocols
- Emergence of new uses

Challenges

- Censorship Versus Moderation
- Scalability



Wait… what about AI?

AI & Hypercentralisation

So... Does size matters?

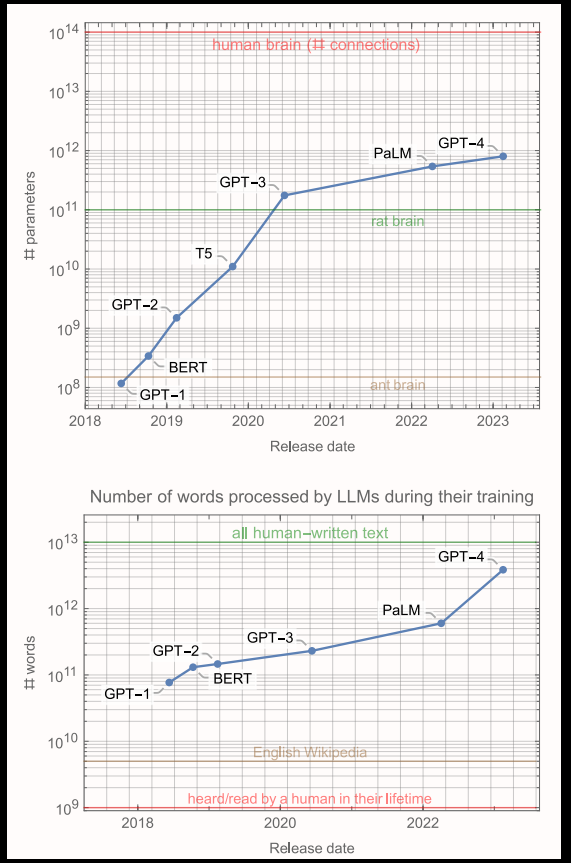
- Bigger Datasets
- More LLM Parameters
- More Resource consumption
- → Moore's Law (x10)

OpenAI → Closed AI

- Business Logic
- Private black boxes
- FUD returns

Shovel sellers during the Gold Rush...

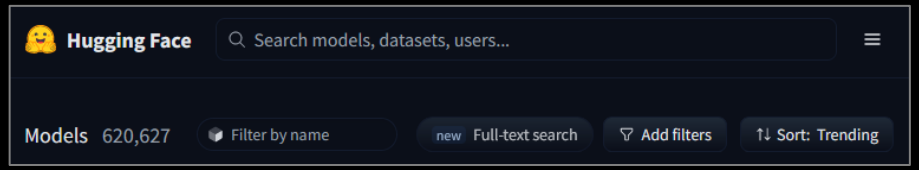
- Hardware Manufacturers
- Cloud Providers



Source: numind.ai

Toward decentralized AI

Openness & Reuse
- Opensource dynamism and collaboration
- Beware of licenses and restrictions



Democratization
- Smaller, more specialized models
- Quantization: on a PC or in a container



OpenAssistant has finished!

OpenAssistant collected data from over 13'000 humans and released it to the public. Data, models, and code are publicly available.

Interoperability
- Formats: OpenAI APIs, ONNX, GGML/GGUF
- Collaborative: Hugging Face, Shoggoth Protocol

Decentralization mechanisms
- Crowd Training: OpenAssistant
- Blockchain and AI: BitTensor



re-Decentralizing



Value



Data



Exchanges



Intelligence

Conclusion



To go further (with me)



Why should we say Internet3 rather than Web3? Let's demystify...



Introducing Web3



Web3 Unveiled: How Decentralization is shaping the future of the Internet

19 Min To Read · 01 Aug, 2023 · Emerging technologies & trends

This article aims to introduce the concept of Web3, sometimes presented as the next evolution of the Internet.

Read more



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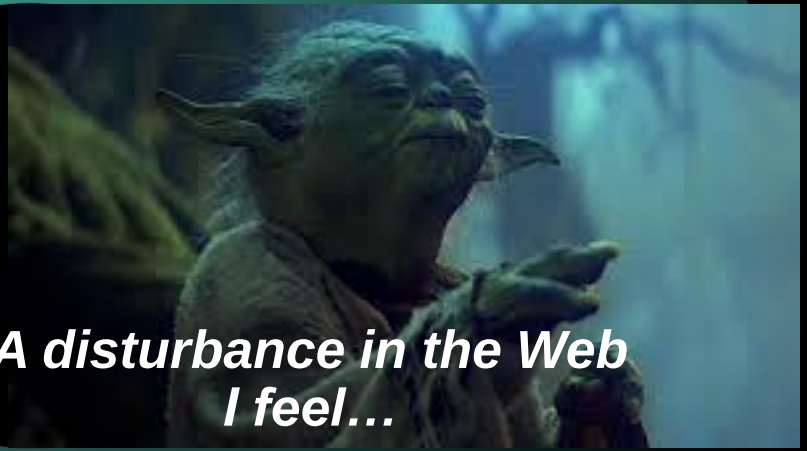
How Open is Generative AI? Part 1

18 Min To Read · 25 Oct, 2023 · Artificial intelligence and data science

Welcome to this two-part series on Generative AI Openness, where we explore the history, current landscape, and potential future of open collaboration and proprietary control in the development of Language Language Models (LLMs).

Read more

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Open Source Expert



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